SPACE ROBOTICS LABORATORY · TOHOKU UNIVERSITY · PHASE 0

Moonwalk 2.0

Report 03 · From one block to a settlement

Report 02 took the site to the pavement. This report takes the pavement to the structures: the block grows into wall, column and dome, then into power, habitats and fluid shells for the programmes of a settlement



TOWARDS A PERMANENT [illegible] ON THE MOON

The SPACE ROBOTICS LAB

SPACE ROBOTICS LABORATORY
TOHOKU UNIVERSITY · SENDAI

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TOWARDS REPORT 04

LIVE REPORT
arhimatrix.github.io/arhimatrix-moonwalk-2-rep_03

INTERACTIVE PARTS
3D Moon tour with site data · block to dome explorer · five end-effectors · scroll sequences

01



SHACKLETON–DE GERLACHE RIDGE · LROC · NASA/GSFC/ARIZONA STATE UNIVERSITY

CHAPTER 01 · MOONWALK 2.0 · REPORT 03

The site

Connecting Ridge, between Shackleton and de Gerlache

Within 150 km of the pole the crater rims catch the low Sun while their floors never see it. The structures in this report are drawn on Connecting Ridge: long sunlit periods, a shadowed floor within reach, and a shoulder flat enough for the pad.

01 · ORIENTATION

Turning to the south pole

The Moon's axis is tilted only about 1.5°. At the pole the Sun never climbs more than a few degrees: light arrives sideways, high ground can stay lit for most of the year, and low ground can stay dark for ever.



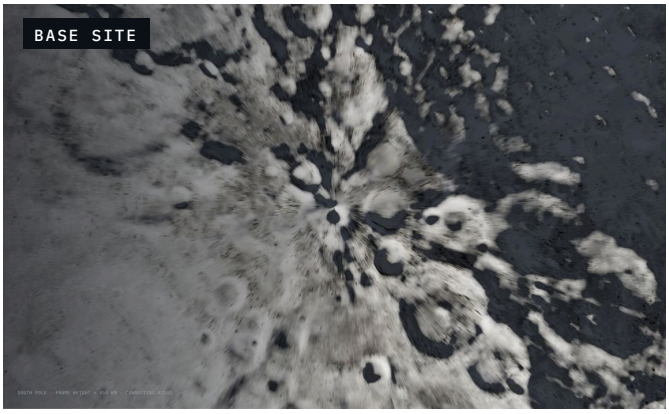
THE 3D MOON OF THE LIVE REPORT TURNS TO THE SOUTH POLE

01 · THE SITE

Connecting Ridge

Report 02 scored four NASA candidate regions; the structures in this report are drawn on Connecting Ridge, between Shackleton and de Gerlache: long sunlit periods, a shadowed floor within reach, and a shoulder flat enough for the pad.

The region can be opened in JSMARS, a browser-based planetary GIS by Prof. Anwar Sounny-Slitine: jsmars.sounny.com, lat −89.46°, lon −99°.



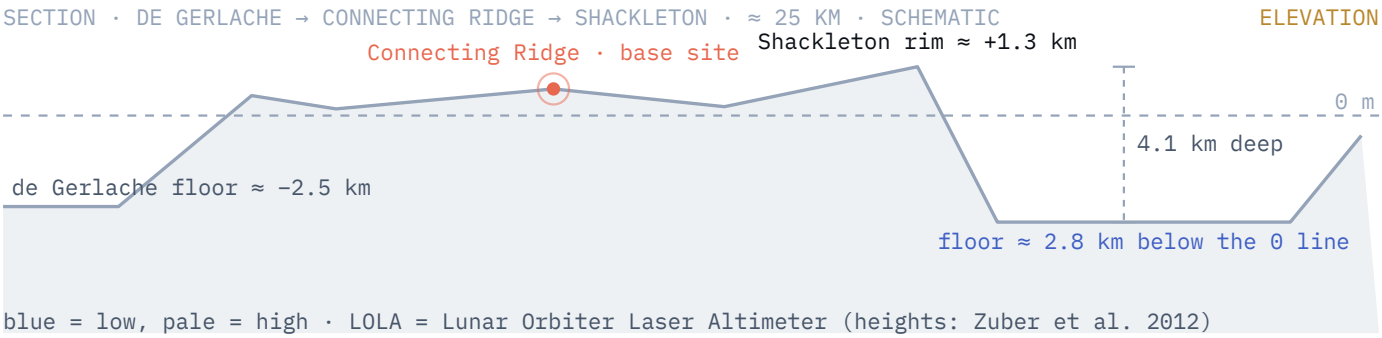
BASE SITE ON CONNECTING RIDGE

01 · SITE DATA

Height, light and cold

Three NASA data sets explain the choice of the ridge, drawn here on one schematic section from de Gerlache across Connecting Ridge to Shackleton.

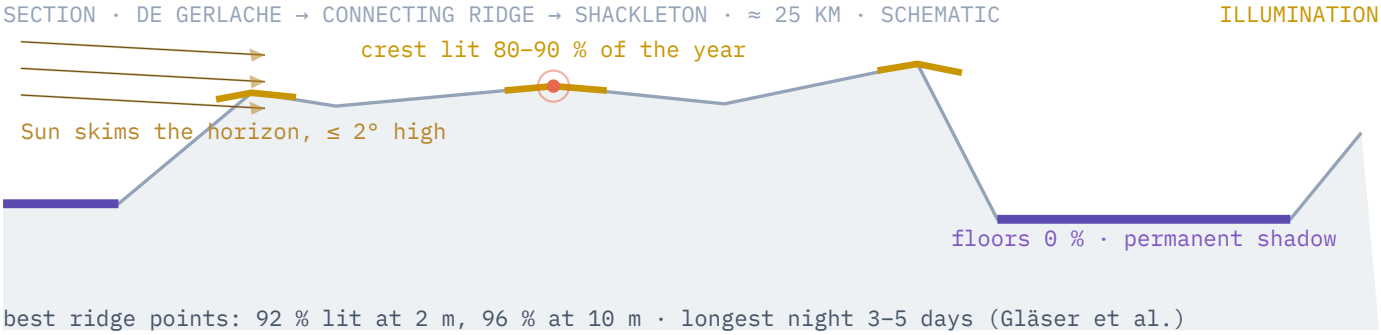
SECTION · DE GERLACHE → CONNECTING RIDGE → SHACKLETON · ≈ 25 KM · SCHEMATIC



Elevation · LOLA

The base site sits on the high crest between the two craters, because at the pole height buys sunlight. Shackleton's floor lies about 4.1 km below its rim.

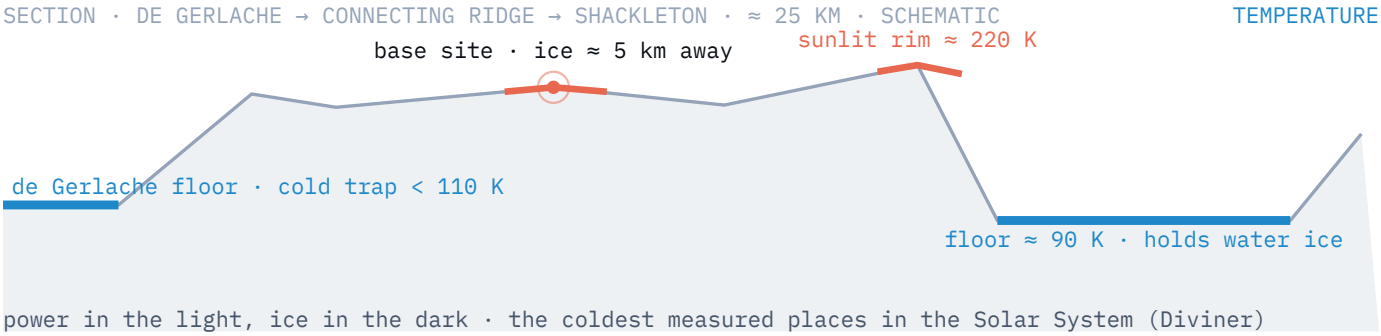
Report layer: LOLA 5 m polar DEM (Zuber et al. 2012).



Illumination

The Sun skims the horizon at no more than about 2°. The crest is lit 80–90 % of the year; the best ridge points 92 % at 2 m and 96 % at 10 m above the ground, with the longest night 3–5 days.

Report layer: Mazarico average-illumination map; values after Gläser et al.

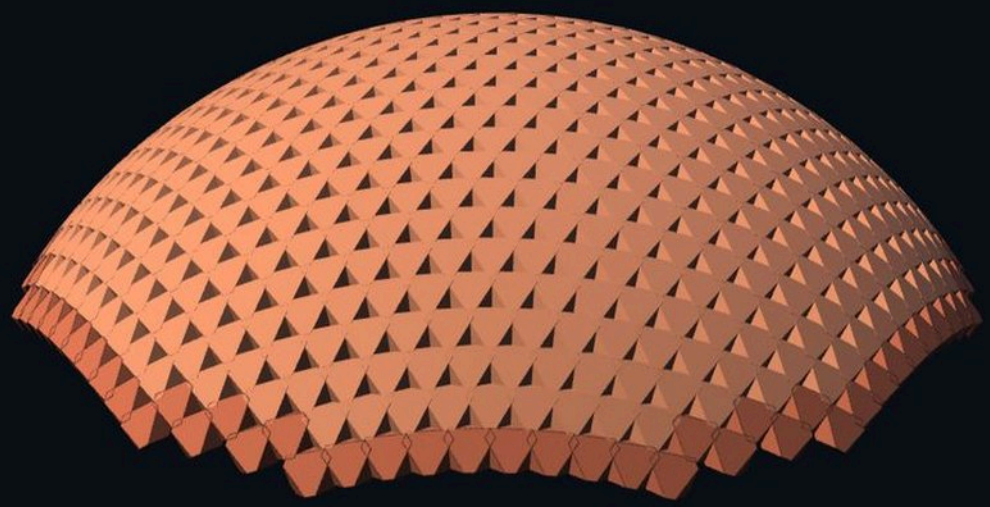


Temperature · Diviner

Floors that never see the Sun stay below 110 K; Shackleton's floor, at about 90 K, holds water ice about 5 km from the base site. The sunlit rim reaches about 220 K.

Report layer: Diviner maximum-temperature map.

02



BLOCK → DOME EXPLORER · LIVE REPORT

CHAPTER 02 · MOONWALK 2.0 · REPORT 03

From block to dome

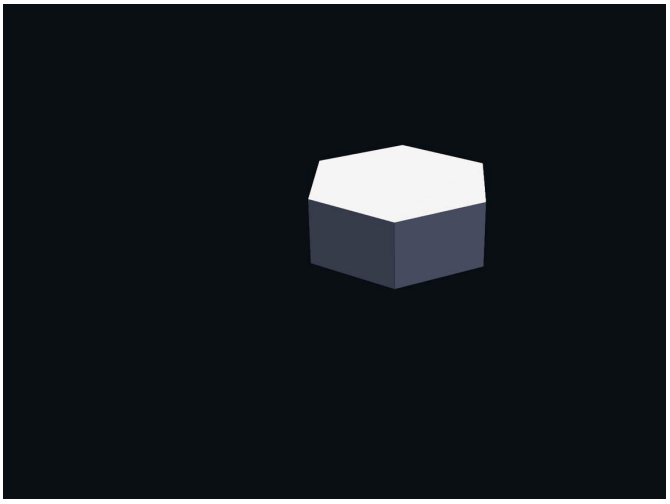
One hexagon, five structures

The block grows into pavement, wall, column and dome. The tilt of its six side faces is the only thing that changes, and it is what makes the blocks lock.

02 · THE BLOCK

One hexagon grows into every structure

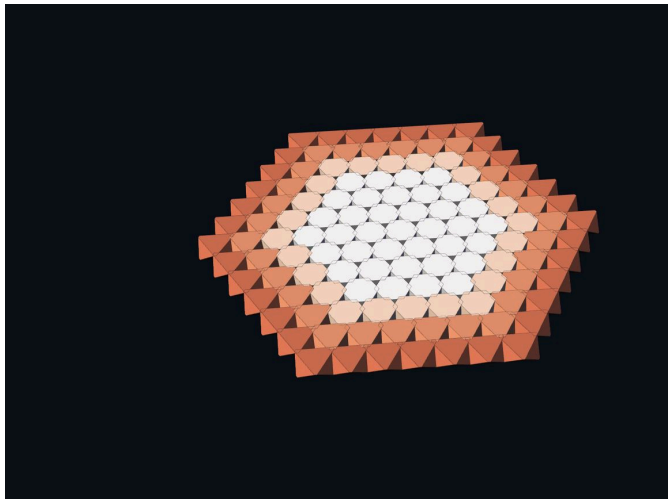
The live report's explorer steps through five structures built from the same hexagonal tilt block. The tilt, thickness, rings and courses can be changed in the browser; the views below show each stage at its default settings.



STAGE 1 · THE BLOCK

One hexagon. One number.

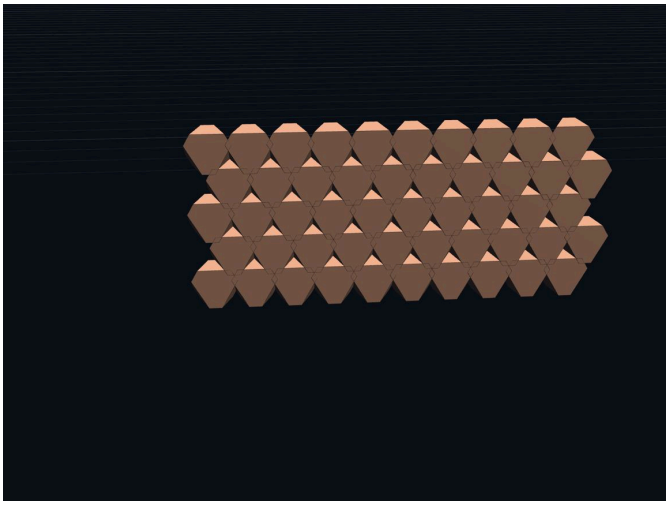
Six side faces that lean alternately in and out. At 0° it is a plain prism; at 35.3° its faces are those of a cube on its corner. 200 mm cube edge, 283 mm across, 100 mm thick, about 12 kg. The tilt is the only thing that changes, and it is what makes the blocks lock.



STAGE 2 · PAVEMENT

Rings from the centre, tilt towards the edge.

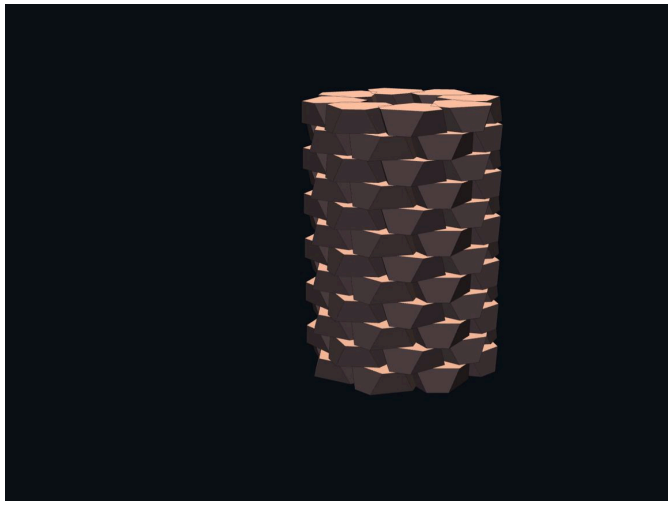
A flat 0° core bedded on compacted ground, locked in by tilted rings whose parted corners open the dust pockets. In Isaac Sim the full-tilt pad stood with only its rim held.



STAGE 3 · WALL

Stand the pavement up.

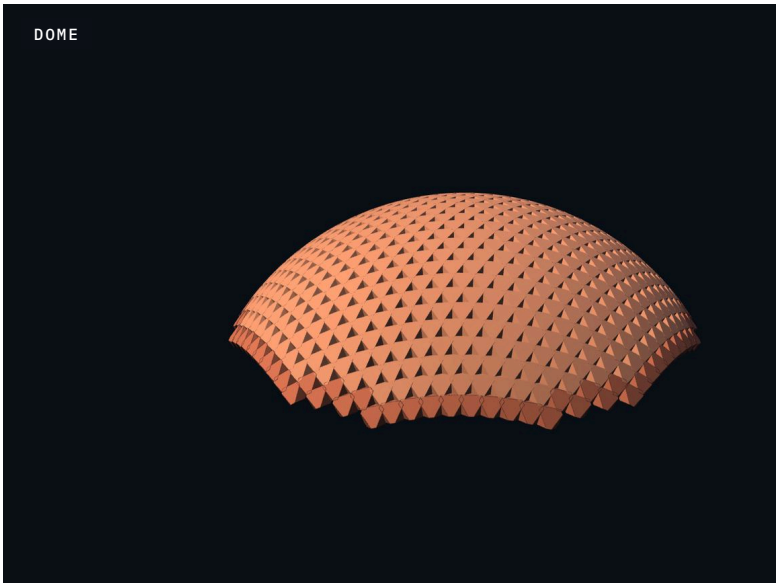
The same block on edge, in courses. Alternating tilts lock each course into the one below, and the pitch equals the pavement pitch, so the wall lands on the grid without cut blocks.



STAGE 4 · COLUMN

Close the wall into a ring.

A hollow hexagonal ring, course on course, filled with loose regolith. Column A stands 2.5 m in Isaac Sim at 10.9 t. Keep stacking and it becomes the power tower.



STAGE 5 · THE DOME

STAGE 5 · DOME

Bend the pavement into a cap.

The pad mapped onto a sphere. Each block leans on its neighbours as it does on the ground, and a rim ring seats the shell, so it carries itself as it grows, with no centering.

STAGE	SIDE TILT	WHAT IT SHOWS
Block	0–35.3°	One hexagonal cell, 200 mm cube edge, 100 mm thick, about 12 kg
Pavement	0° core, 35.3° edge	Graded rings; dust pockets open where the tilted corners part
Wall	19.9°	Courses lock into the one below; pitch equals the pavement pitch
Column	15°	Hollow ring filled with regolith; 2.5 m, 10.9 t in Isaac Sim
Dome	25°	Pad mapped onto a sphere, seated on a rim ring, built without centering

03



Drag to orbit · + / − or pinch to zoom · three blocks at 17.6° tilt, the middle one is picked · in option 1 it is shown translucent to reveal the cavity

− +

OPTION 3 · THREE-FINGER POCKET GRIP · LIVE REPORT MODEL

CHAPTER 03 · MOONWALK 2.0 · REPORT 03

Robot interface

How the robot holds the block

The hexagonal tilt block changes the gripping problem: every block has the same flat top, and the tilted side faces alternate. Five ways to grip it, on the same blocks, through the same four steps.

03 · ROBOT INTERFACE

How the robot holds the block

Report 01 compared end-effectors on the osteomorphic block. Here: Report 01's twist-key and flush side handle carried over, and three grips that use the new geometry. Three blocks at 17.6° tilt; the middle one is picked.

OPTION 1 · TOP TWIST-LOCK

Key into the top slot, turn 90°

A slot is printed into the flat top over a round cavity. The T-key drops through the slot, turns 90° inside the cavity and bears up under the slot lip: a form lock, no friction needed. The same key fits every block in the family because the top is the same on all five tilts.

Approach **vertical**
Actuation **wrist turn 90°**
Block feature **slot + cavity**
Hold **form lock**

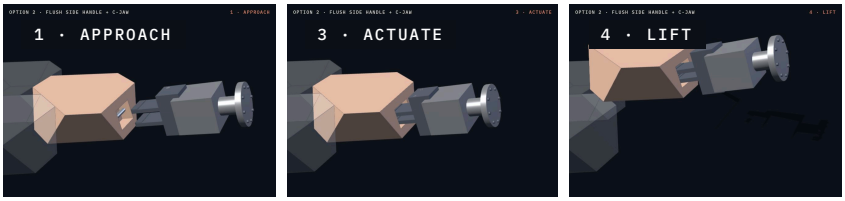


OPTION 2 · LAB-COMPATIBLE

Flush side handle + C-jaw

Report 01's baseline on the new block: a 12 mm steel bar across a 70 × 56 mm pocket in one side face, so nothing sticks out of the block. The lab's existing C-jaw approaches square to that face, reaches into the pocket and closes on the bar from above and below. One face must be free; the other five can be in contact.

Approach **horizontal**
Actuation **jaw closes**
Block feature **recessed handle**
Hardware **existing**



OPTION 3 · TOP POCKET GRIP

Three fingers into printed top pockets

Three triangular pockets are printed into the flat top, one at each short edge. Three fingers on radial slides drop in from above and spread outward until their flat sides bear on the outer pocket walls; the hold is symmetric, so the block hangs level, and the pockets are the same on every tilt.

Approach **vertical**
Actuation **fingers spread**
Block feature **3 top pockets**
Hold **form + friction**



OPTION 4 · MAGNETIC TOP PLATE

Electromagnet on the flat top

No vacuum on the Moon, so no suction. A hexagonal plate with an electromagnet seats on the flat top; the block carries a thin ferrous insert under its top face, printed in. Fastest cycle, lowest precision needed, and the plate doubles as the pocket-magnet charger.

Approach **vertical**
Actuation **coil on**
Block feature **ferrous insert**
Hold **magnetic**



OPTION 5 · THREE-FINGER SIDE GRIP

Self-centring on alternating faces

Three fingers from above, landing on the three faces that tilt inward. Because those faces lean the same way on every block, the fingers centre the block as they close. Needs the neighbouring cells empty on three sides, so it suits placing, not retrieving from a finished course.

Approach **vertical**
Actuation **fingers close**
Block feature **none**
Hold **wedge + friction**



Why the top face wins

Every block in the family shares the flat top, so one top-gripping effector serves pavement, wall, column and shell. Side faces change tilt from ring to ring; a side jaw would need five settings.

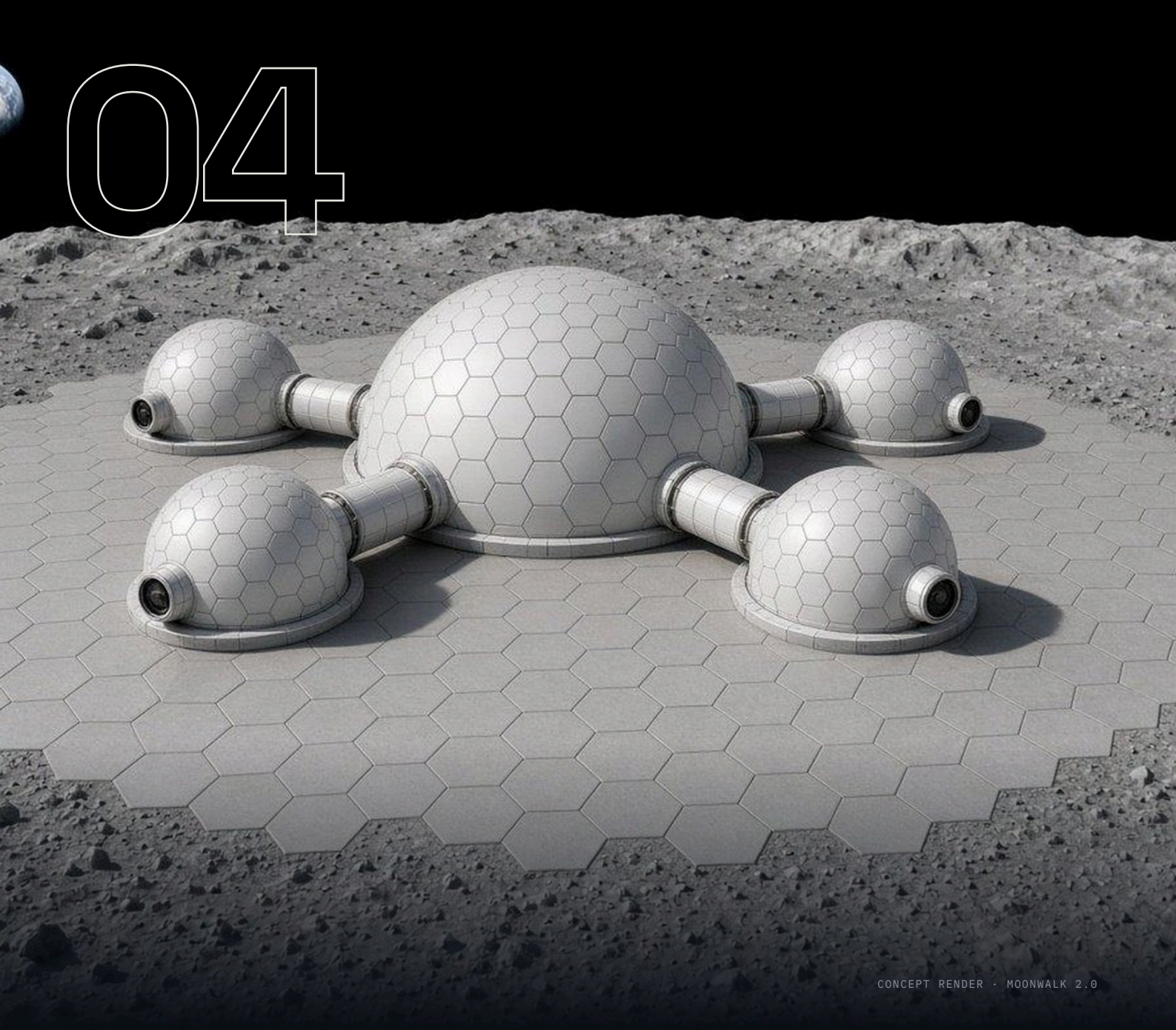
Why the pockets help

Three shallow triangular seats printed at the short top edges, 26 mm deep, with 20 mm of material left to the face. They are the same on all five tilts, so one three-finger hand picks every block in the family.

What stays lab-compatible

Option 2 runs on the lab's C-jaw as it is and needs one free side face. The other four grip from above; the twist-key slot and the finger pockets are printed into the top, the same on every tilt.

04



CONCEPT RENDER · MOONWALK 2.0

CHAPTER 04 · MOONWALK 2.0 · REPORT 03

Habitats

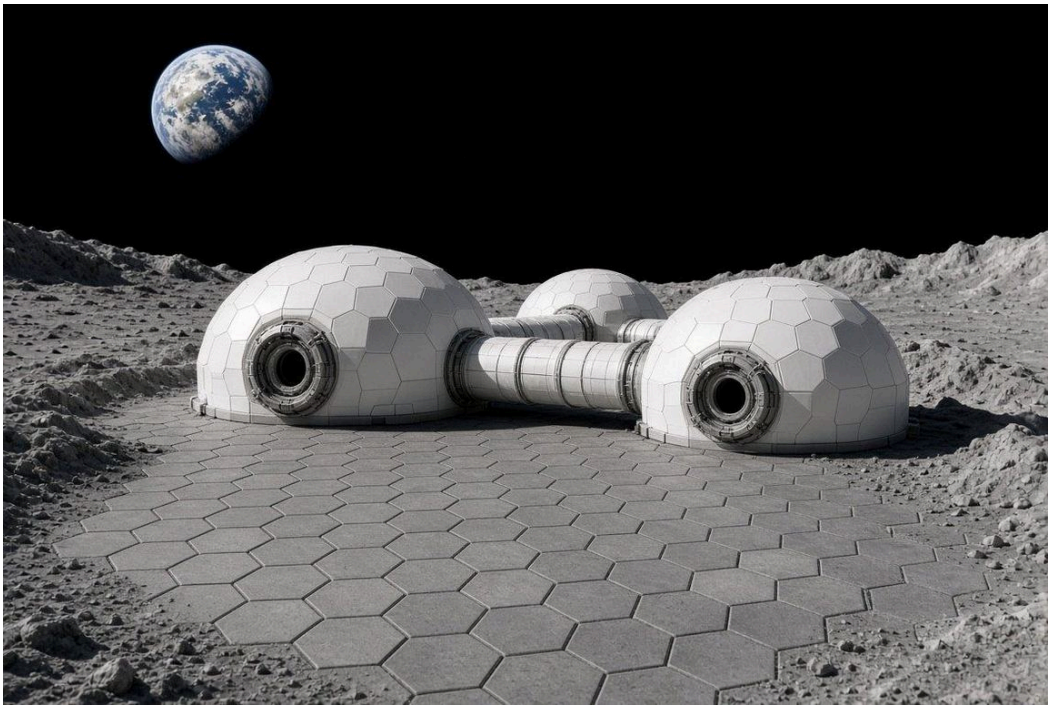
Habitats grow out of the pavement

A levelled seat in the paving becomes the floor of a dome, and the same block closes it over. Domes dock in chains, clusters and rings, and berms of regolith add the shielding.

04 · HABITATS

From two domes to a ring

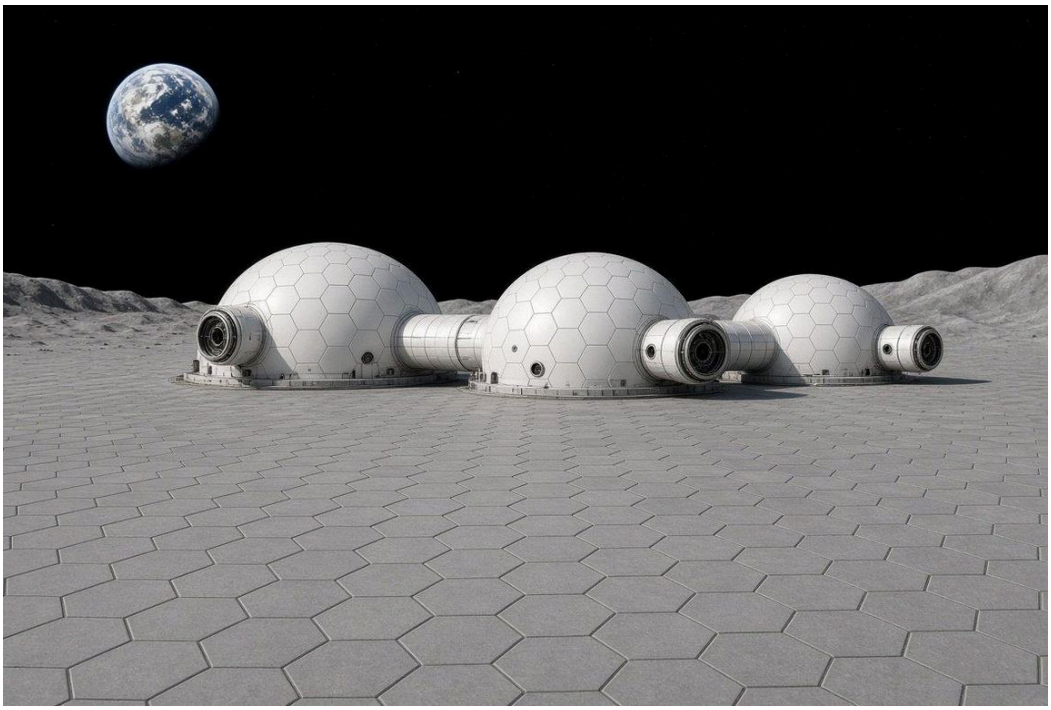
Five steps of growth, each on its own paved pad. Concept renders: the sky and the Earth in them are illustrative; from the south pole the Earth sits low on the horizon.



04 · TWO DOMES AND A TUNNEL

Habitats grow out of the pavement.

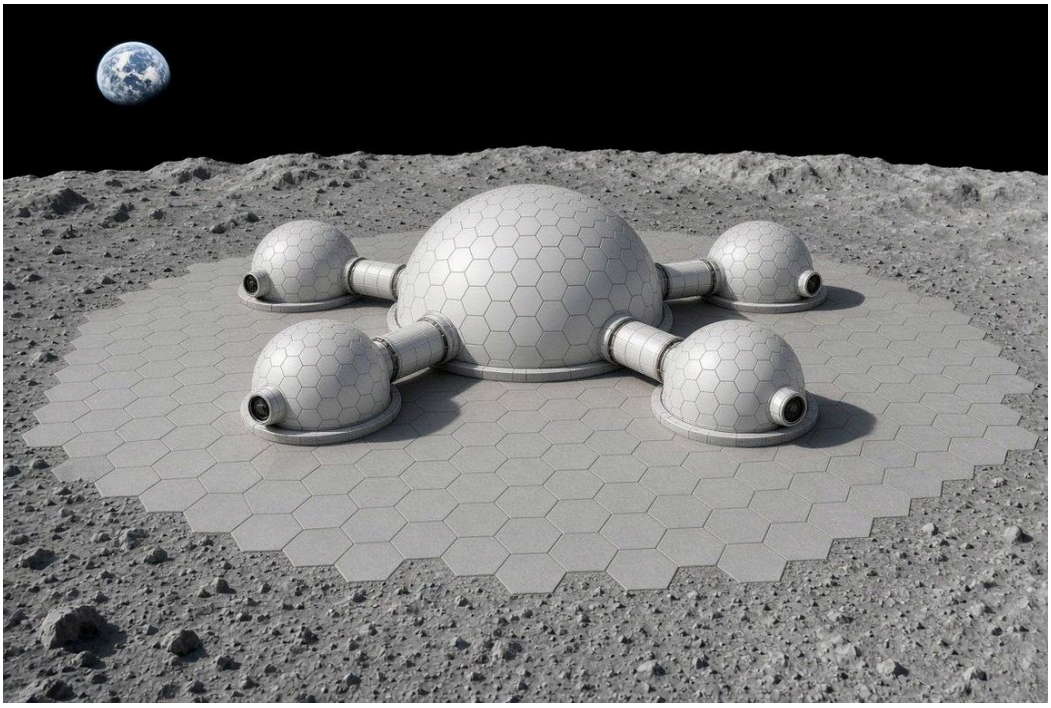
The shell is the pavement bent into a cap: a levelled seat in the paving becomes the floor of a dome, and the same block closes it over. Two domes and a tunnel are the first habitable unit.



04 · THREE IN A LINE

Three in a line.

Elements dock in a chain, the way NASA plans Multi-Purpose Habitat, Node and Airlock. Each port is the 2.0 m ring planned for surface mating; logistics ports are rated for 60 dockings over ten years, crew ports for three.



04 · FIVE

One, then five.

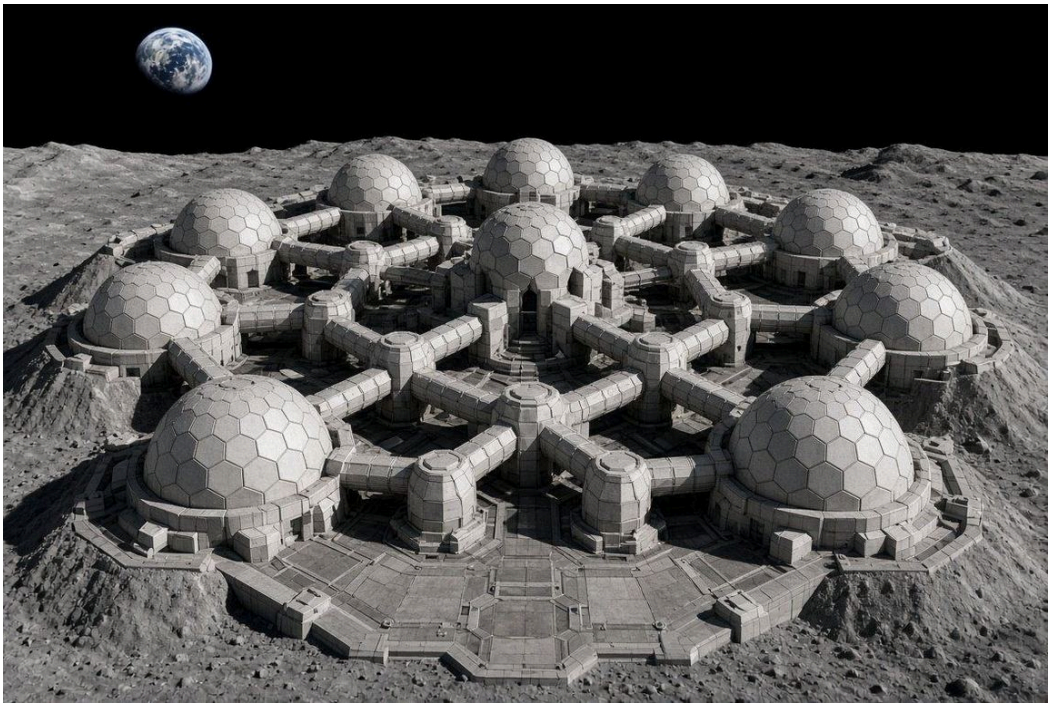
A central dome and four satellites on short tunnels: the smallest modular cluster, on its own paved pad.



04 · BURIED

Under the regolith.

Berms of loose regolith over the shells add the shielding mass. From outside only the ports and the paved aprons show.



04 · RING

The ring.

Clusters join into rings and the base grows one ring at a time. Every port gets a flat apron; corridors stay below 20° for vehicles.

05



CONCEPT RENDER · MOONWALK 2.0

CHAPTER 05 · MOONWALK 2.0 · REPORT 03

Inside the habitats

Soft walls, hard shell

The pressurised volume is delivered, not printed: an inflatable liner inside the block vault. The block does the shielding; the soft structure does the living.

05 · INSIDE

Common room, corridor, work bay, quarters

8.2 psi and 34 % oxygen past the port. The interiors follow NASA's surface habitat figures for translation paths and docking ports.



05 · COMMON ROOM

Soft walls, hard shell.

Past the port it is 8.2 psi and 34 % oxygen. The pressurised volume is delivered, not printed: an inflatable liner inside the block vault. The block does the shielding; the soft structure does the living.



05 · CORRIDOR

The corridor.

A 1.1 × 2.09 m translation path stays clear through every hatch, enough for a suited crew member in a contingency.



05 · WORK BAY

Work bay.

Benches, racks and bunks along the service side of each element, reachable from a shirt-sleeve environment. In-situ repair is limited, so maintenance items are few and accessible.



05 · QUARTERS

Quarters.

A private cabin each, tunable light on a 24-hour cycle, and a window to the pole.



05 · SERVICE SIDE

The systems live in the walls.

Air revitalisation, water recovery, thermal control and power distribution run behind the liner, with the trunk from the pavement entering at each port.

06



CONCEPT RENDER · MOONWALK 2.0

CHAPTER 06 · MOONWALK 2.0 · REPORT 03

Power

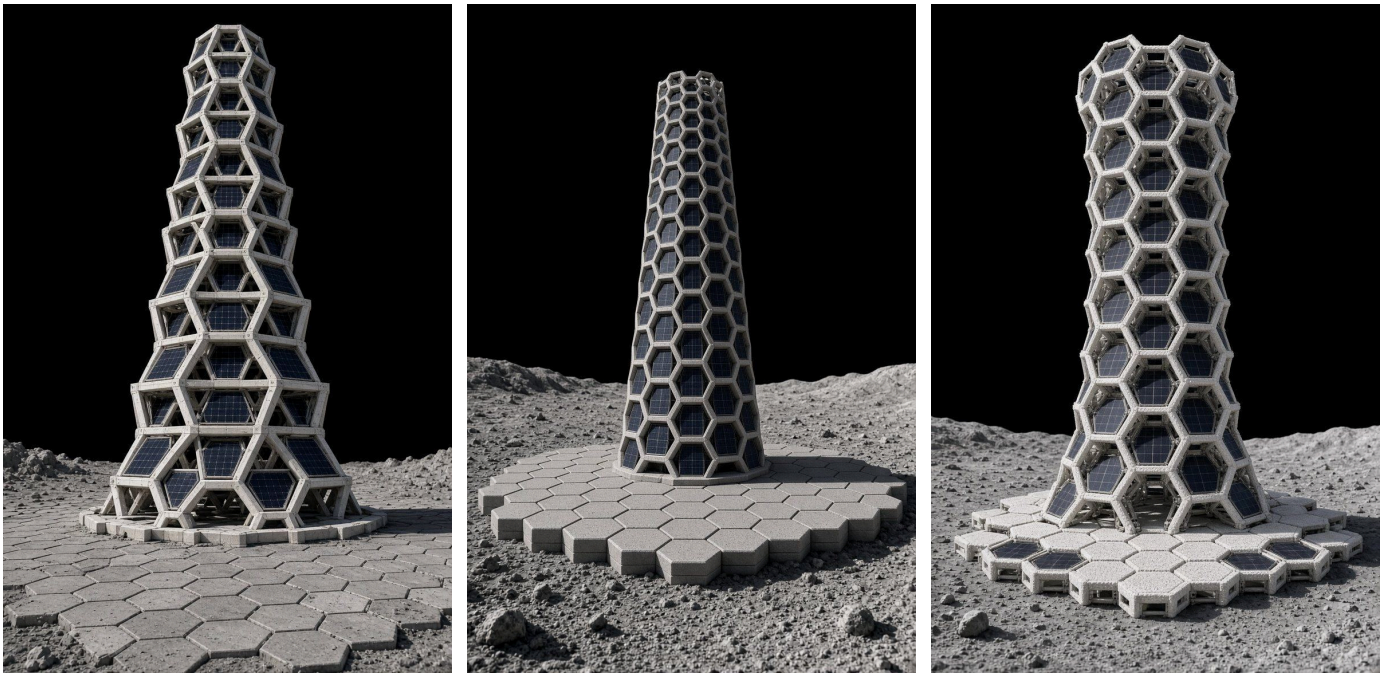
A field of columns is a power plant

Column A keeps growing into a lattice tower. At the pole the Sun stays near the horizon, so panels stand up: each hexagonal cell of the lattice is a seat for one array.

06 · POWER

The column keeps growing

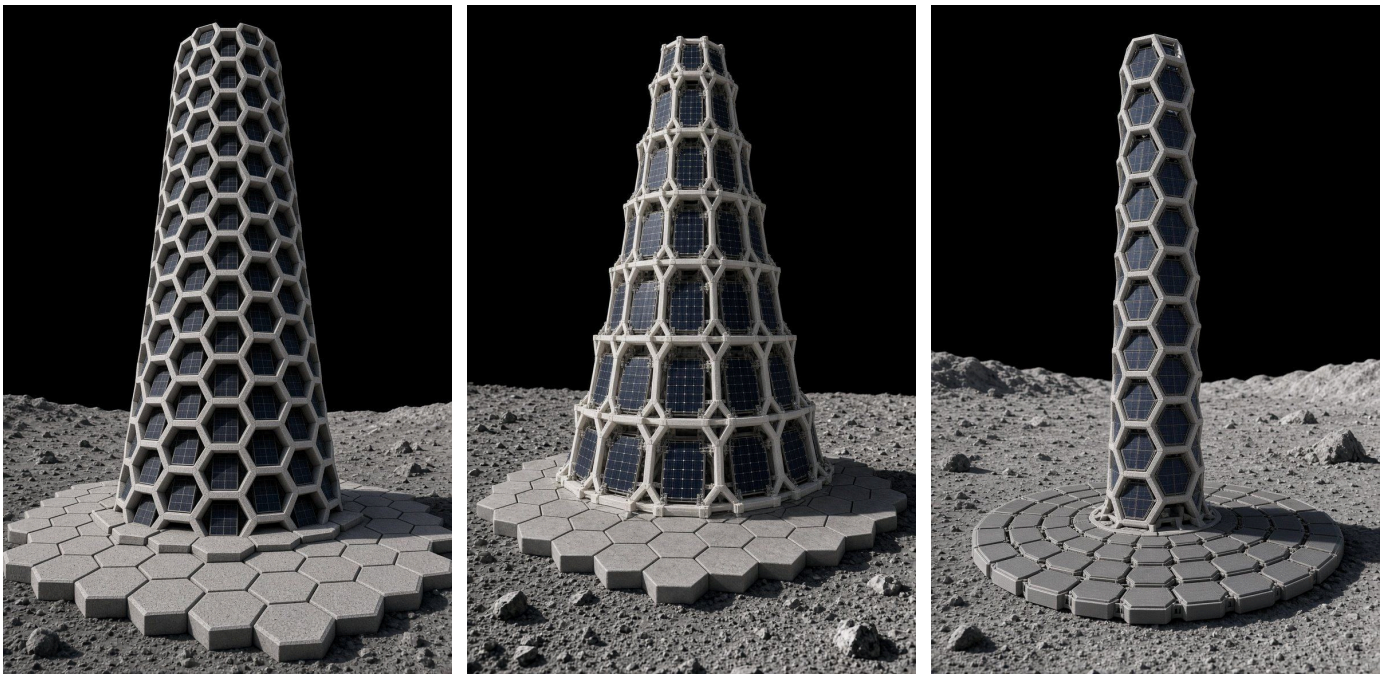
Eighteen towers from the same block, in six steps. Concept renders.



06 · COLUMN → TOWER

The column keeps growing.

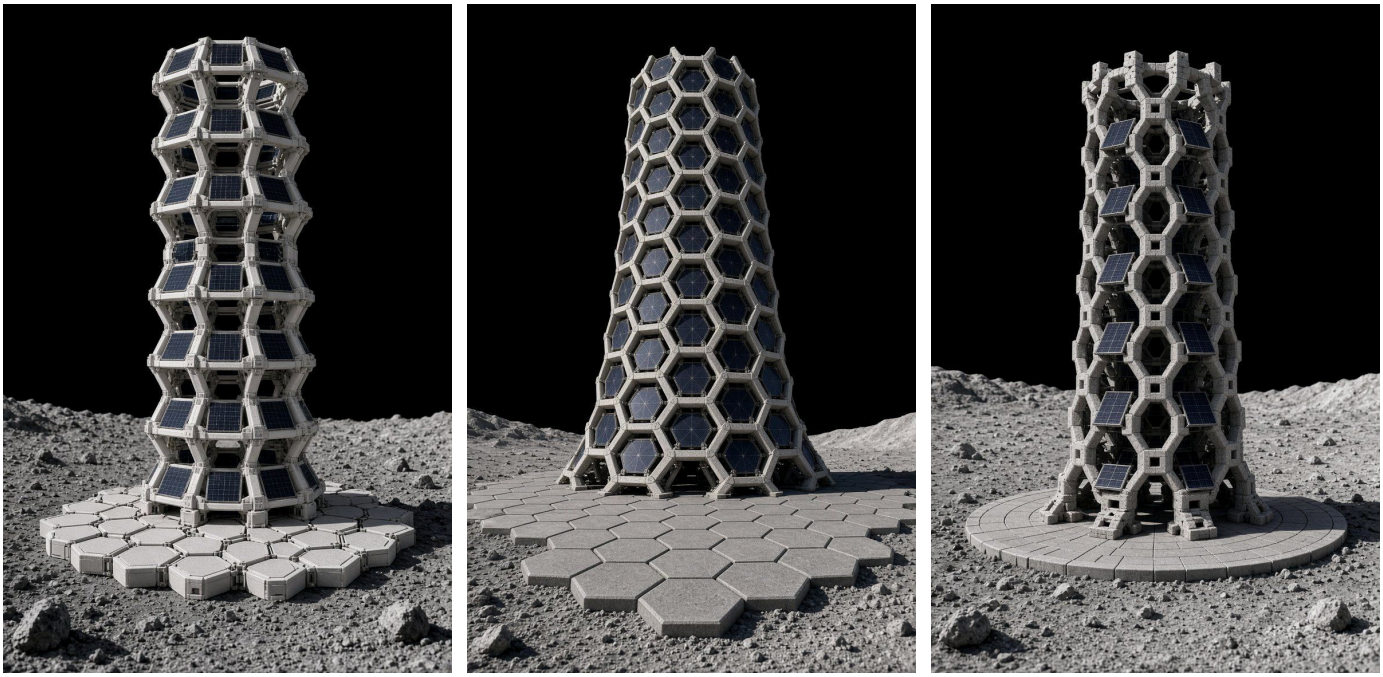
Column A was a hollow ring of blocks 2.5 m high. Drop one cell per course and it tapers; keep stacking and it is a tower. The hexagonal cells are empty, and that is the point.



06 · PANELS IN CELLS

Every cell takes a panel.

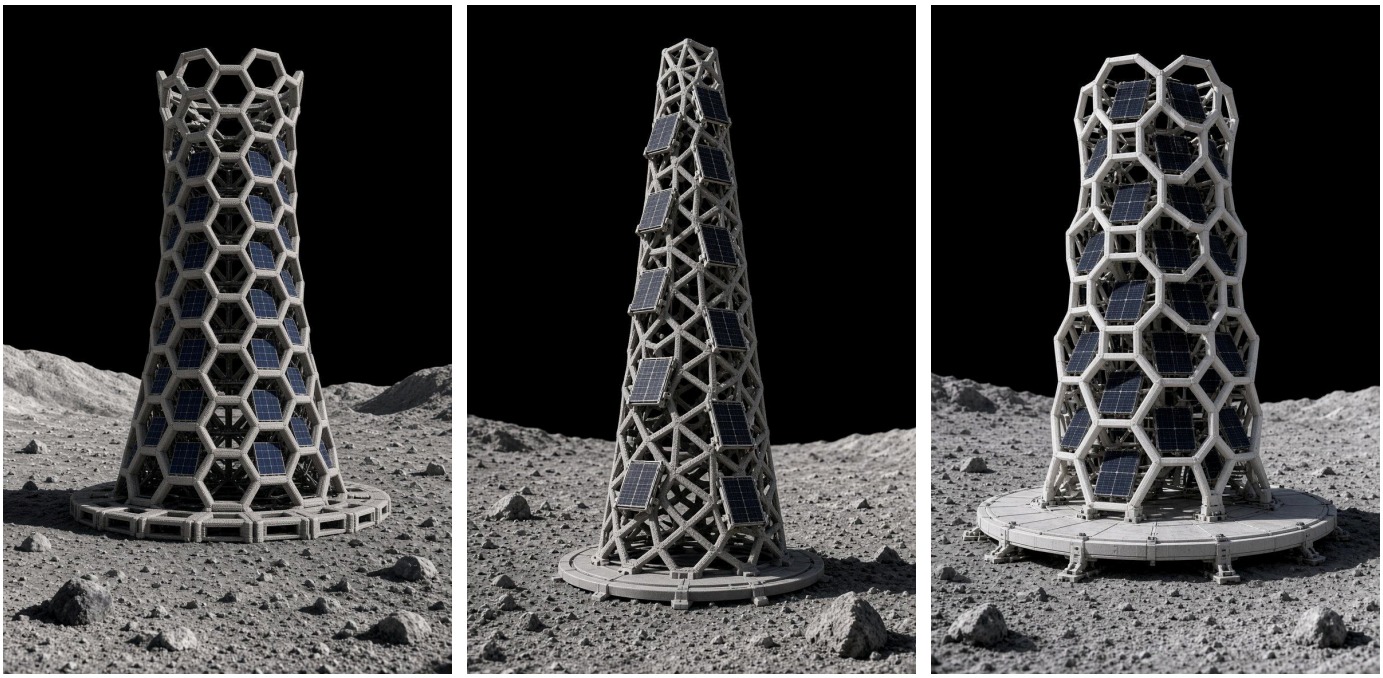
At the pole the Sun stays near the horizon, so panels stand up rather than lie flat. Each cell of the lattice is a seat for one array: the structure is built on site, the panels arrive pre-made and clip in.



06 · OPEN / CLOSED

Open lattice, closed lattice.

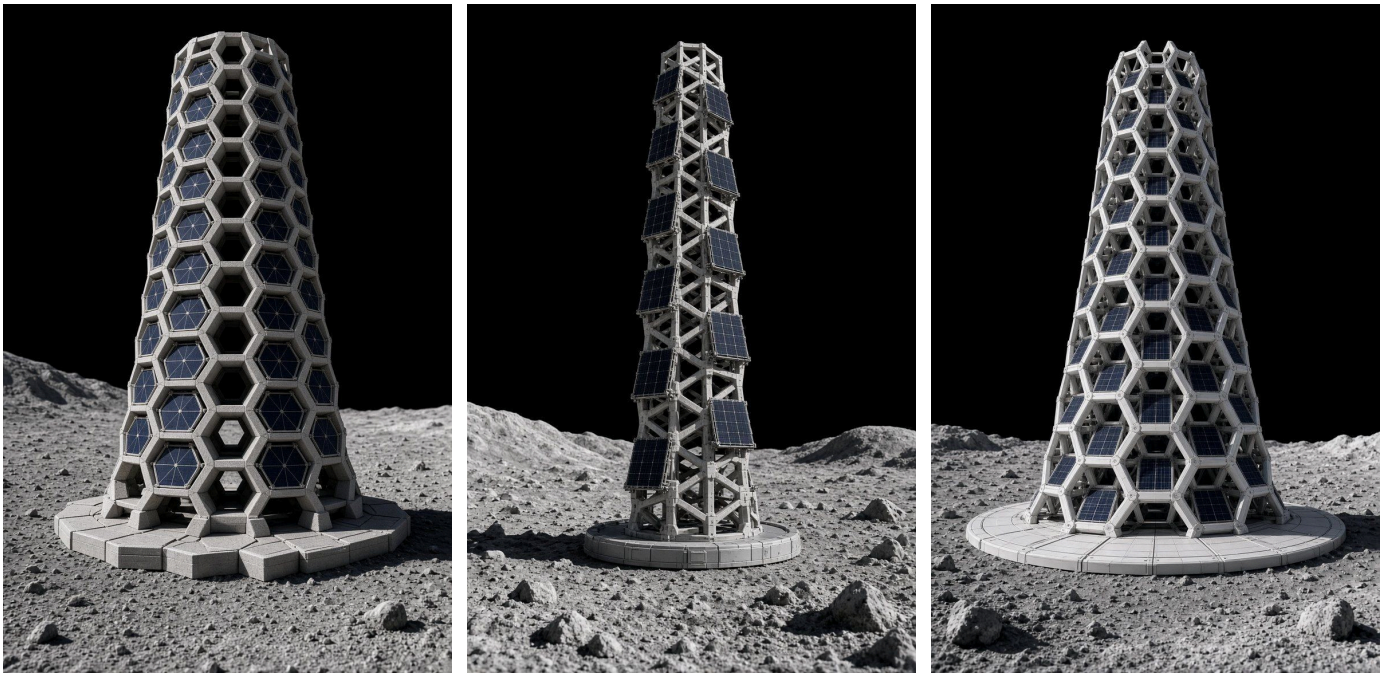
How many cells are filled is a power decision, not a structural one. The same stack can carry a few panels on a sparse frame or a full skin of them.



06 · BLOCK / STRUT

Block or strut.

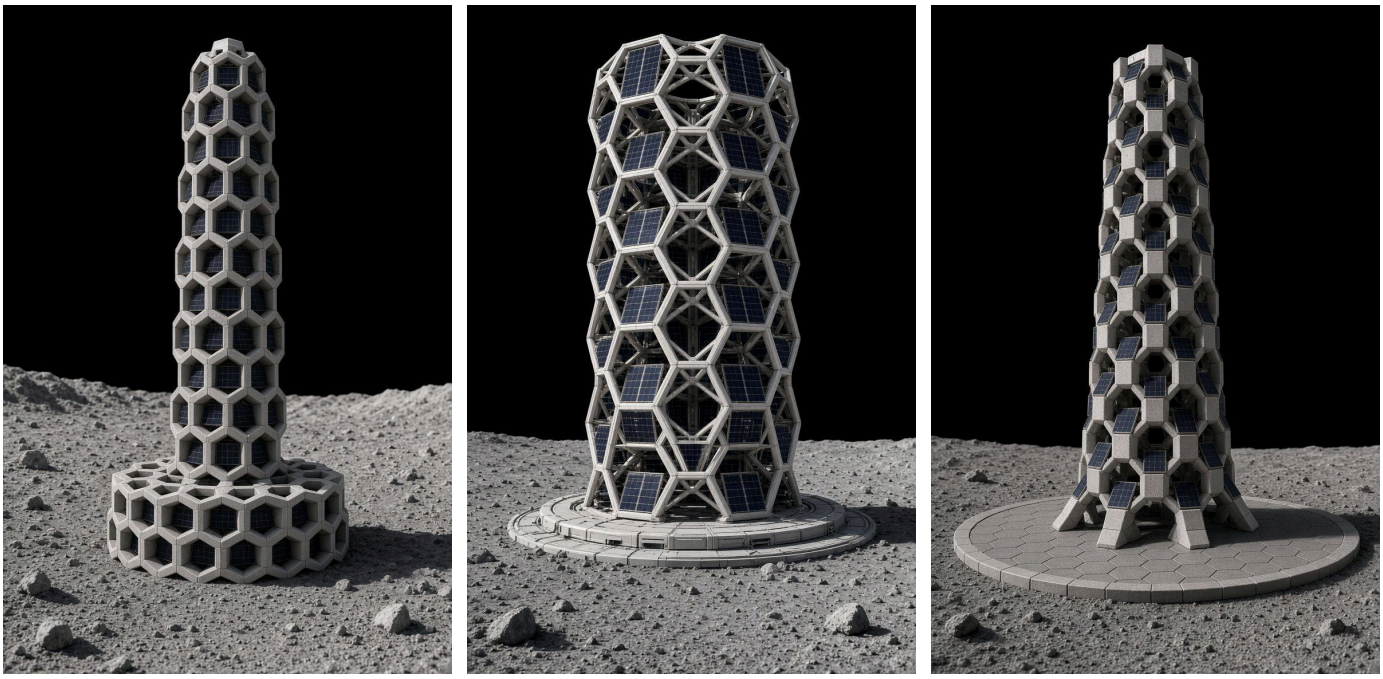
The lattice can be laid as solid block cells or as the block's edges only, a strut frame printed from the same geometry. Both stand on the same paved footing.



06 · TAPER / PLINTH

Tapers and plinths.

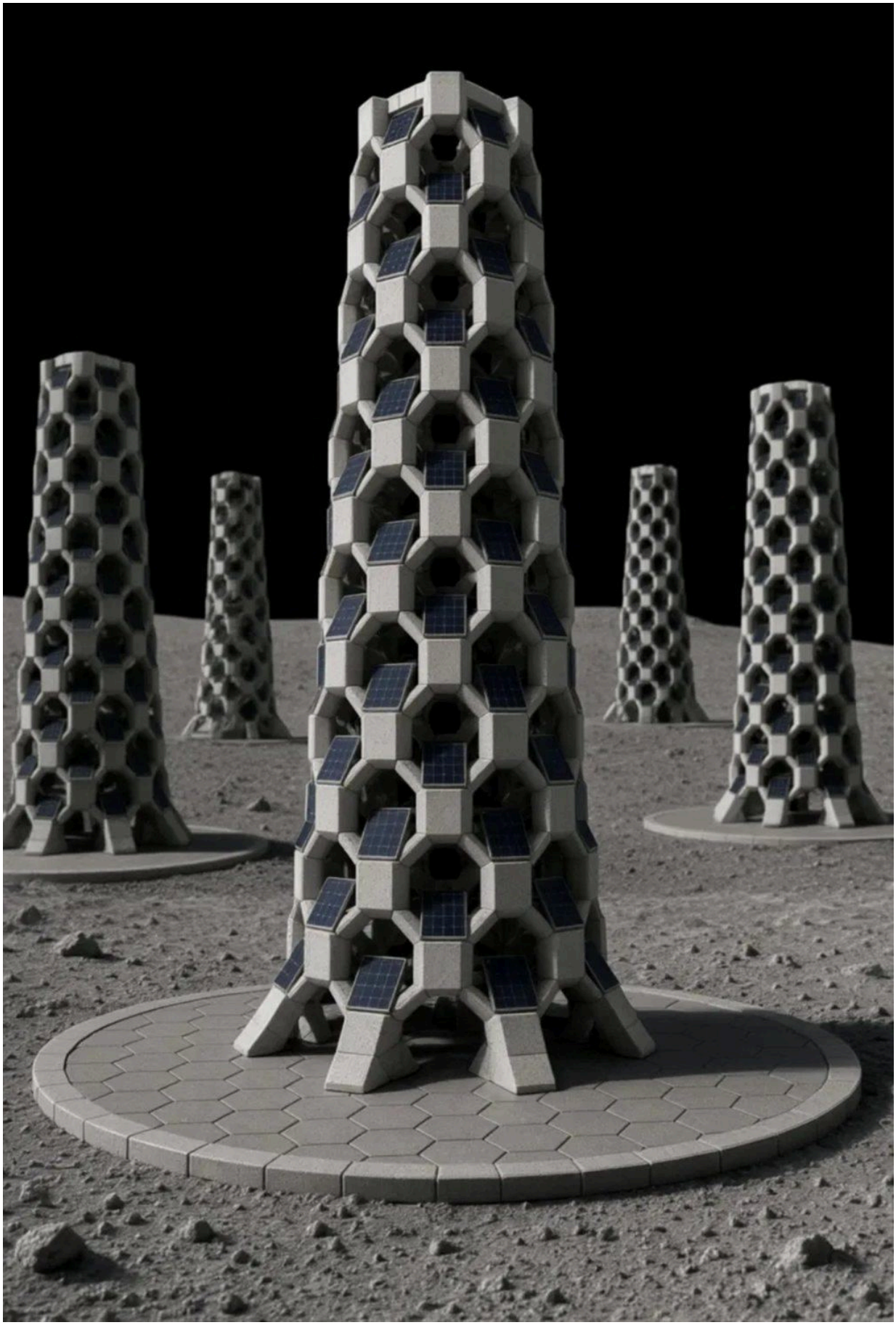
A tower can taper by dropping a cell per course, or rise straight from a wider ring that doubles as a plinth and service base.



06 · VARIANTS

Many answers to one question.

Eighteen towers from the same block: different tapers, densities and panel seats. The lattice is a family, and the right member depends on the site's light and the plant's size.

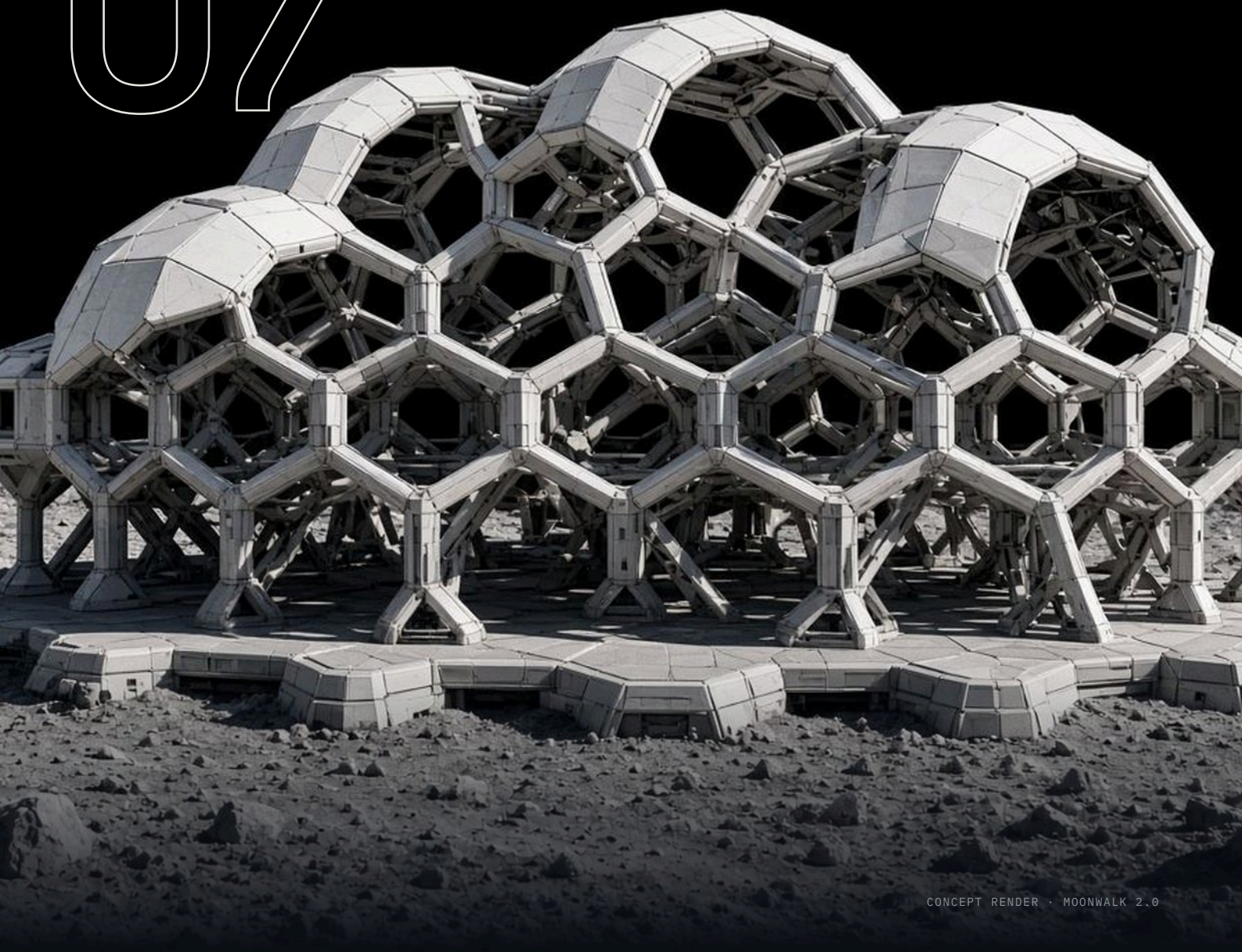


06 · FIELD

A field of columns is a power plant.

Multiply the tower on paved footings and the outpost has its plant. The trunk runs under the pavement to the vault and each bay, 6 kW at 120 VDC per port.

07



CONCEPT RENDER · MOONWALK 2.0

CHAPTER 07 · MOONWALK 2.0 · REPORT 03

Fluid shells

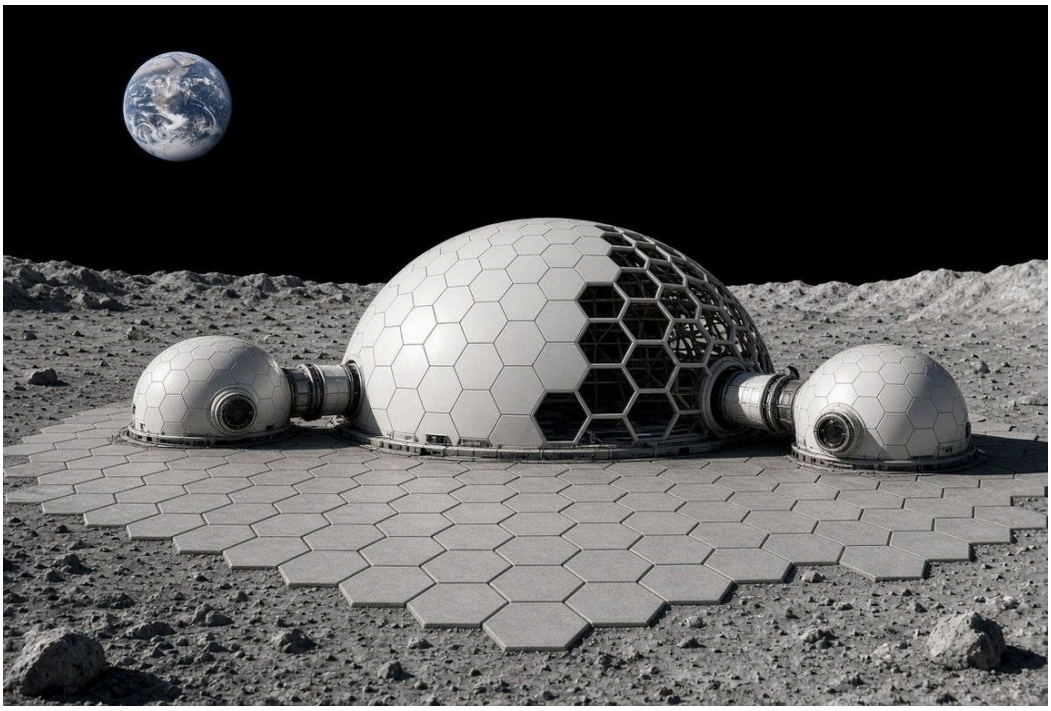
A dome is a choice of which cells to fill

Take the shell apart and the lattice shows: the same block with cells left open. Closed for shielding, open for light, the same frame becomes greenhouse, workshop, auditorium or institute.

07 · FLUID SHELLS

Decompose the dome

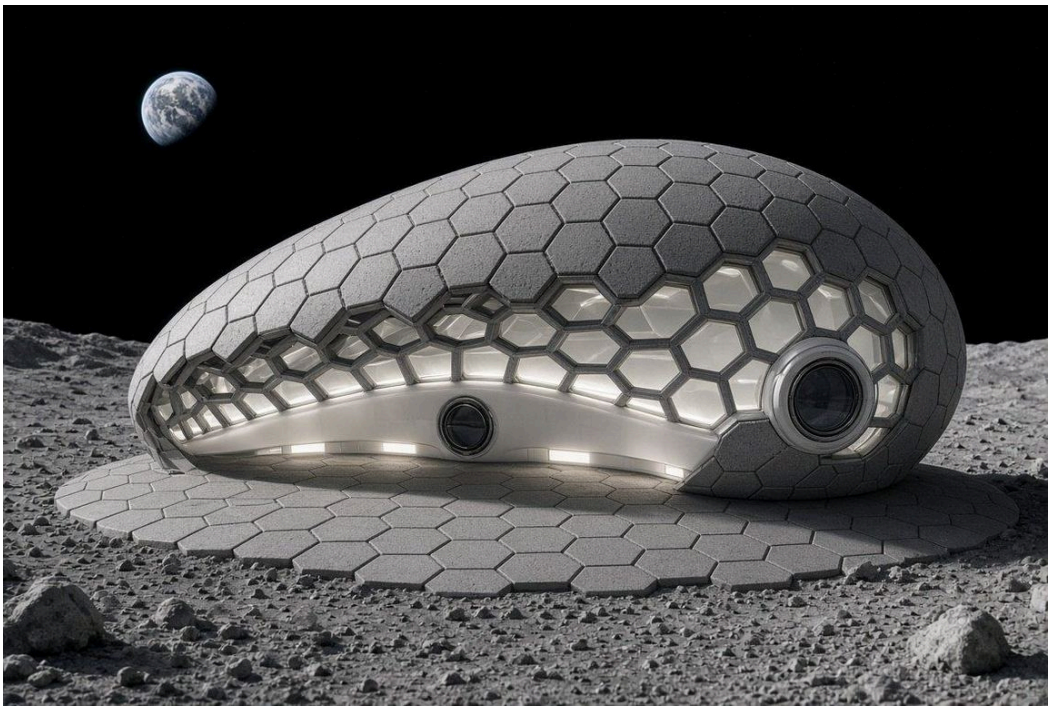
Twelve shells from the same lattice, each a different pattern of filled and empty cells for a different programme. Concept renders.



07 · HALF OPEN

Decompose the dome.

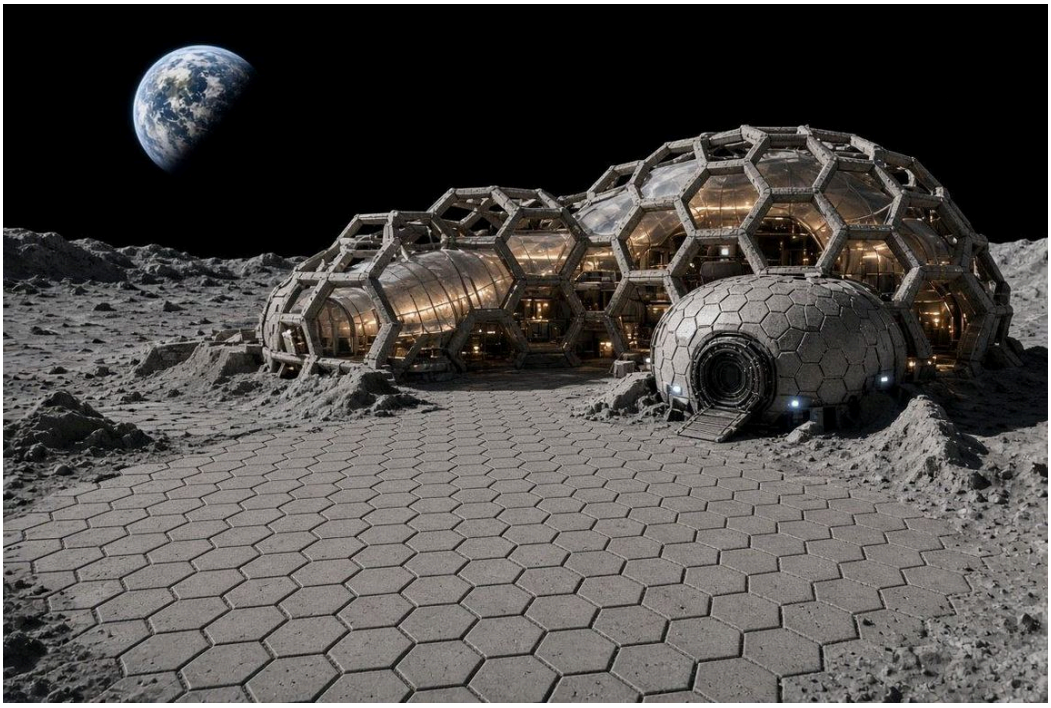
Take the shell apart and the lattice shows: the same block with cells left open. Half closed for shielding, half open for light; a dome is a choice of which cells to fill.



07 · GREENHOUSE

Greenhouse.

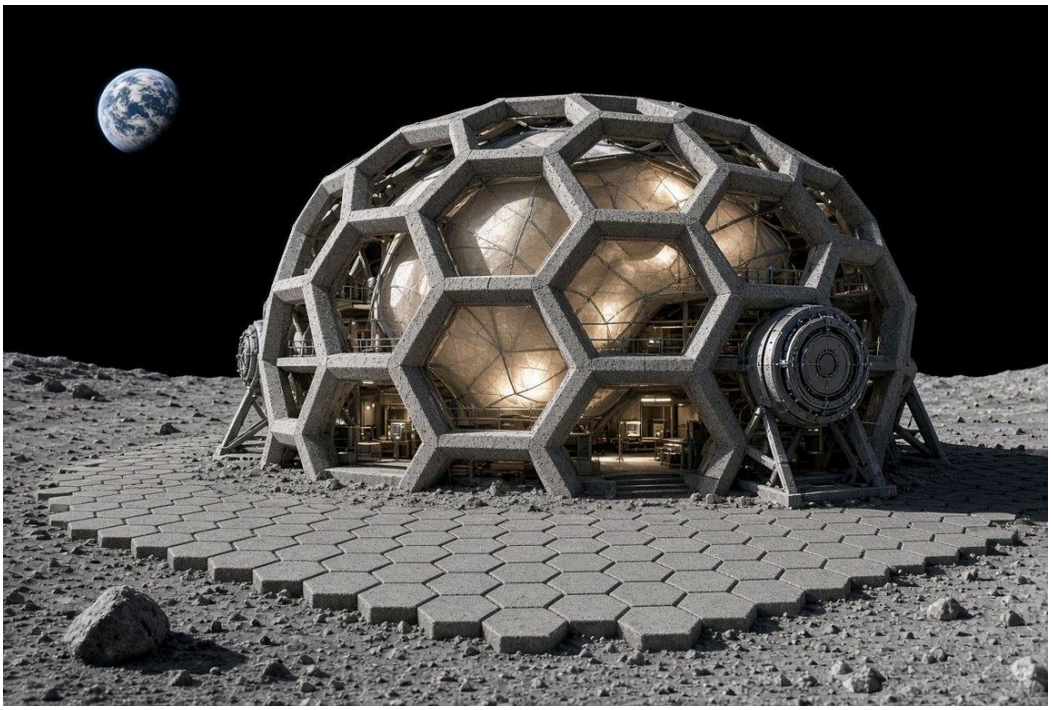
Open the lattice along one side and glaze it, and the shell becomes a greenhouse: fresh food, humidity, oxygen and something green to look at.



07 · WORKSHOP

Robotic workshop.

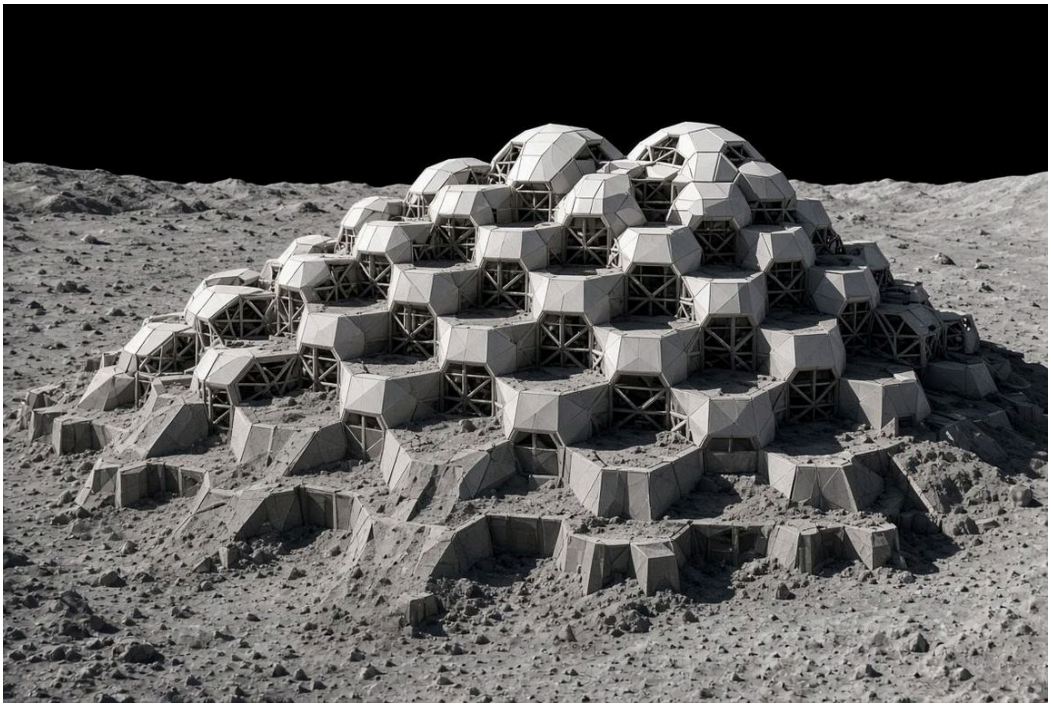
Leave the lattice open and the frame is a hangar. The robot fleet is maintained here, under a roof the same robots built.



07 · LARGE BAY

Robotic workshop, the large bay.

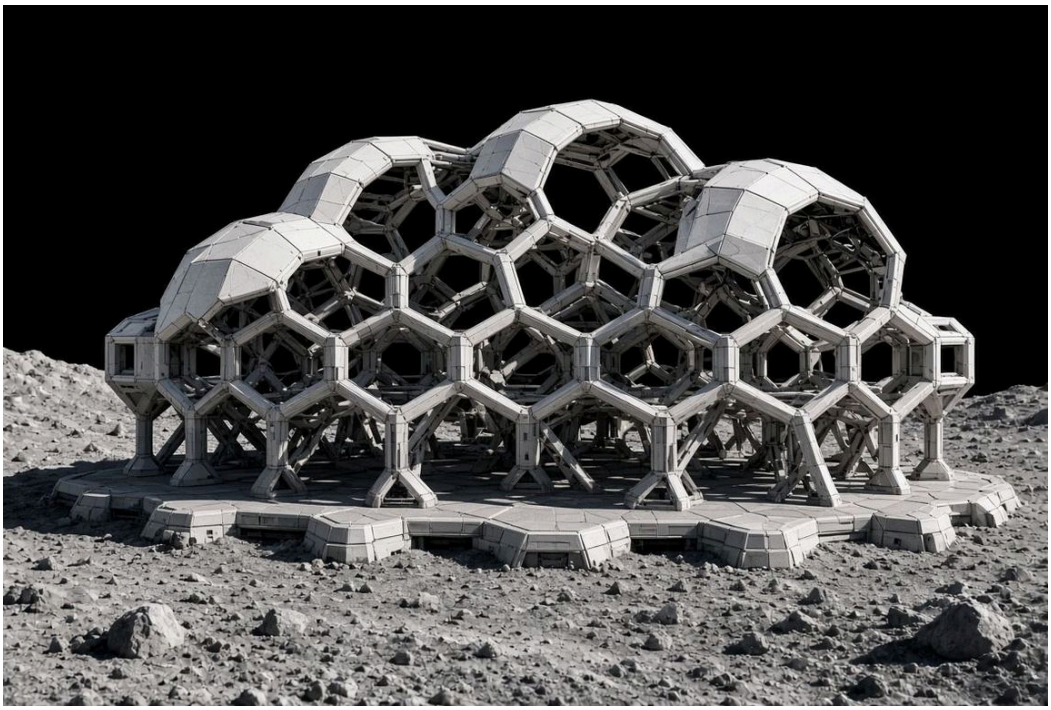
One big cell for the heavy work: block printers, the pavers and the maintainer, with room to lift a habitat element. The frame carries only itself.



07 · STACKED CELLS

Stacked cells.

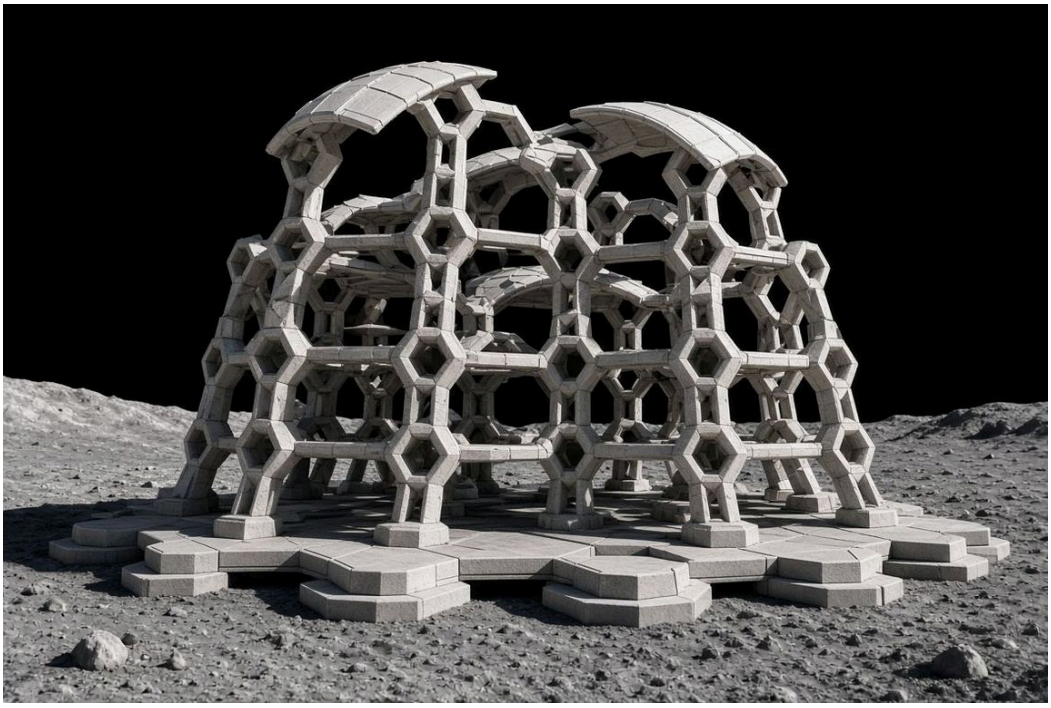
The lattice is not one room but many: cells stacked and offset, each a bay, an alcove or a passage. Variety is built in by where the cells sit, before any interior arrives.



07 · OPEN LATTICE

Open lattice.

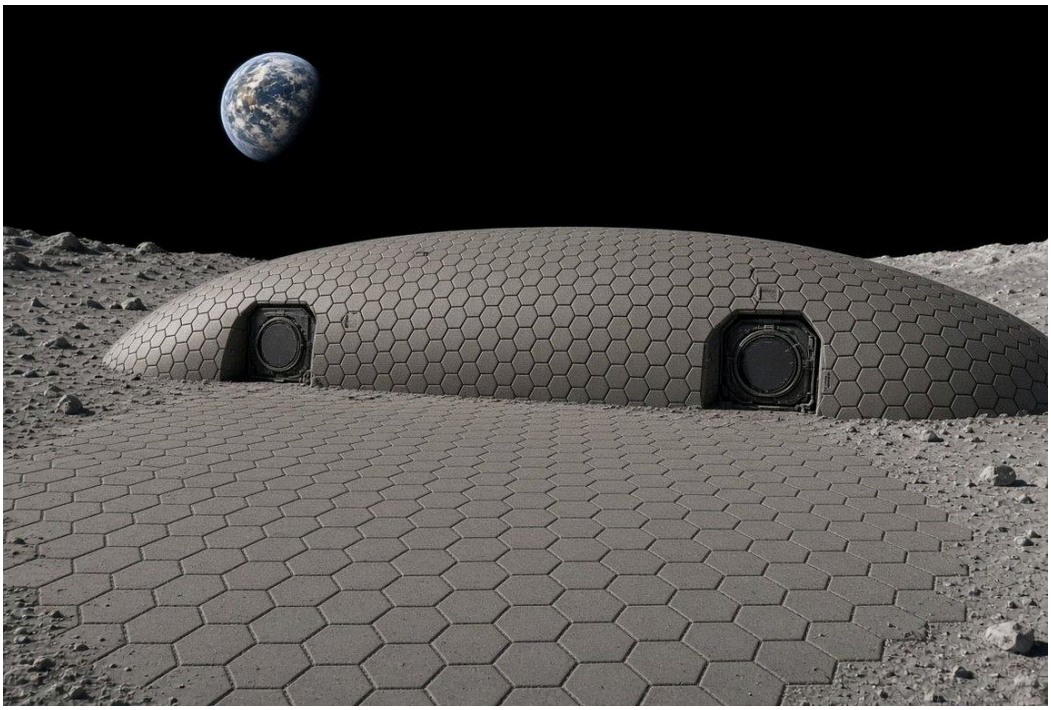
Leave the cells as a frame on the pavement and the structure is a scaffold for rooms: pressurised modules slot into the cells, and the gaps between them become the routes.



07 · FRAME

The frame alone.

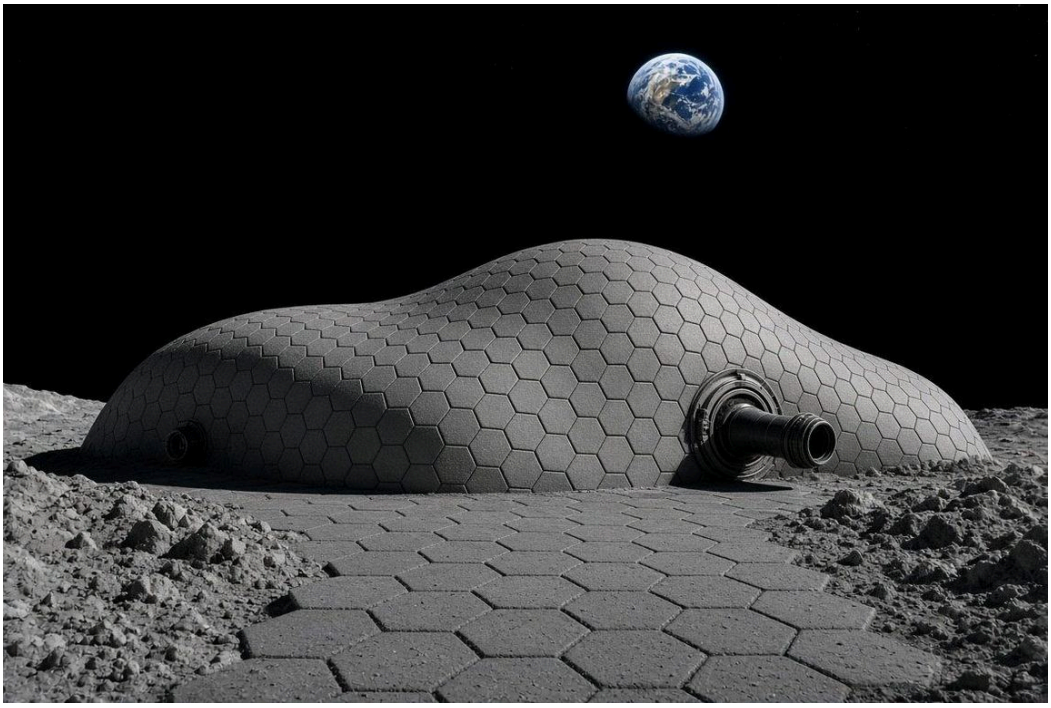
Rings of block stacked into a hexagonal frame, roofed only where it matters. This is the skeleton of every fluid shell: the same unit, the same gripper, a different pattern of filled and empty cells.



07 · CLOSED

Closed and buried.

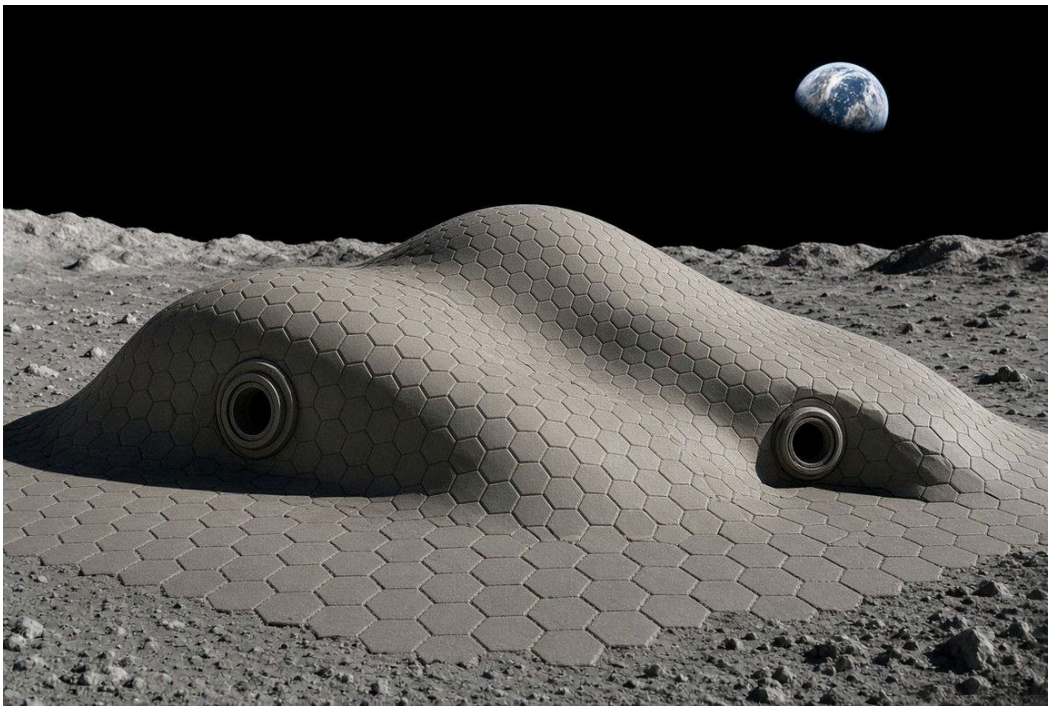
Every cell filled and a berm over it: the radiation shelter, the medical bay, the quiet rooms. One thickened vault everyone can reach quickly.



07 · IMMERSIVE

Immersive section.

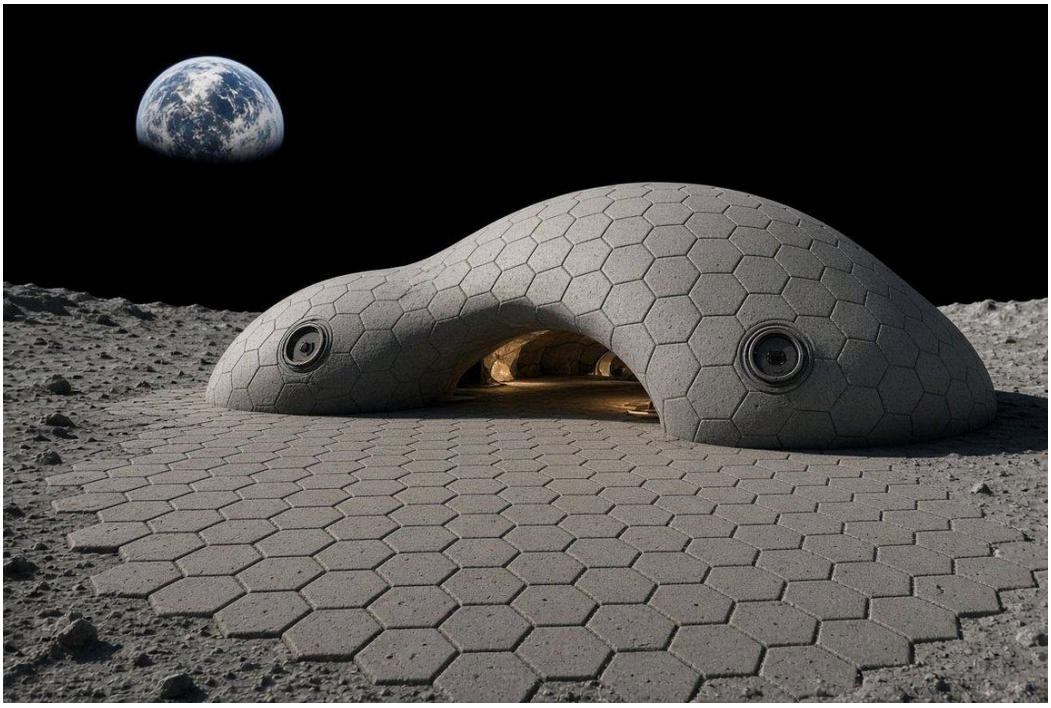
The lattice stretched into a long shell: a walk-through space for projection and sound, the settlement's cinema and gallery in one.



07 · AUDITORIUM

Auditorium.

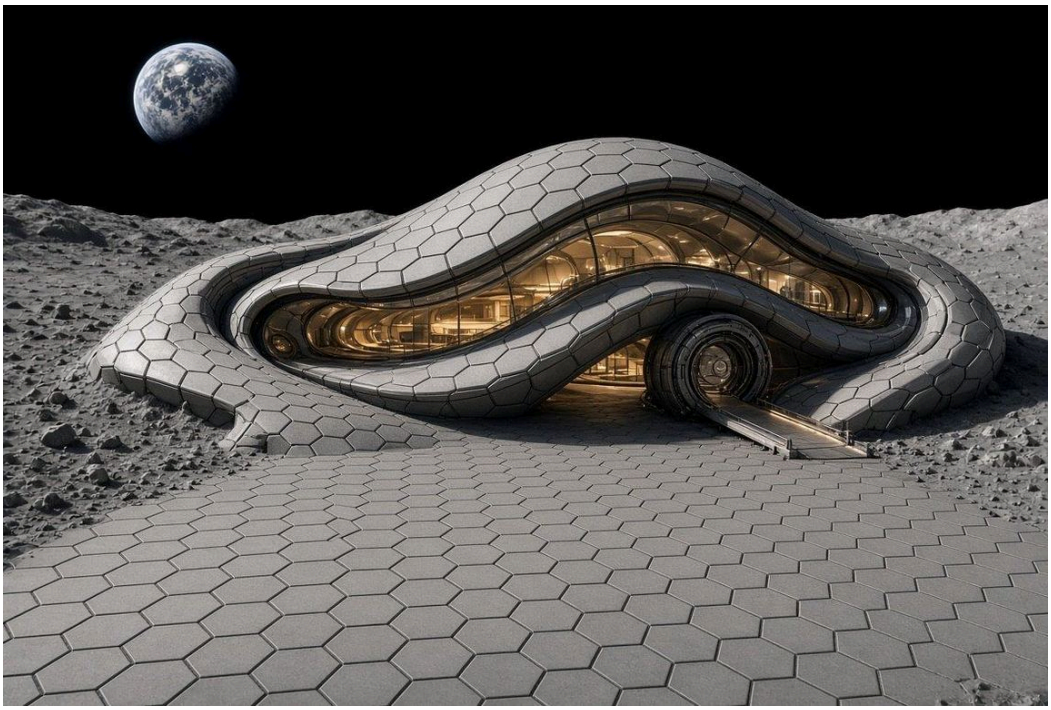
Two volumes under one closed skin: the hall on one side, the foyer on the other, the saddle between them the lobby. Lectures, performances, the town meeting.



07 · INSTITUTE

Arts and science institute.

An arch rather than a vault: studios and labs on both sides, and a covered yard between the ports that opens straight onto the surface, so field research and EVA start at the door.



07 · EVA / GREENHOUSE

Robot and human in EVA.

The ribbon shell glazed where it faces the Earth. A place to work side by side with the robots outside, or a second greenhouse: the lattice does not decide, the programme does.



CONCEPT RENDER · MOONWALK 2.0

CHAPTER 08 · MOONWALK 2.0 · REPORT 03

Inside the shells

Rooms for body, mind and community

Greenhouses, an institute hall, auditoria, immersive rooms, a lounge, a circle and a quiet room: soft structure inside hard shell, with the lattice reading through the liner.

08 · INSIDE THE SHELLS

Soft structure inside hard shell

Sixteen interiors of the fluid shells. The liner carries pressure; the block carries the regolith. Concept renders.



08 · GARDEN ROOM

Greenhouse · the garden room.

Planting beds built into the seating, under a liner that follows the lattice above it. Humidity, oxygen and green in the common space.



08 · GREENHOUSE

Greenhouse · the racks.

Racks of crops under grow light in a long shell. Fresh food, and the loop that keeps the air breathable.



08 · INSTITUTE HALL

Moon Art and Science Institute.

Terraced seating, a stage, and the Moon itself on the wall. Lectures and performances for the settlement; the town hall on other days.



08 · AUDITORIUM

Auditorium.

The closed twin shell from inside: a curved screen at the front, the lattice cells reading through the liner overhead.



08 · AUDITORIUM · LIT CELLS

Auditorium · lit cells.

Raked seating under a ceiling of lit cells: the lattice above, read through fabric.



08 · AUDITORIUM · GATHERING

Auditorium · gathering.

Stepped platforms for a crowd, with the block showing through at the edge where the liner meets the shell.



08 · IMMERSIVE · LUNAR VAULT

Immersive · lunar vault.

A projected Moon on the ceiling: a window where there can be none, during the weeks of a solar event.



08 · IMMERSIVE · CRATER WALL

Immersive · crater wall.

The surface outside brought in at full scale, for people who cannot go out and for those planning to.



08 · IMMERSIVE · OBSERVATORY

Immersive · observatory.

The dome from below, with the far side of the Moon overhead. A room for looking up.



08 · IMMERSIVE · STARFIELD

Immersive · starfield.

Tunable light on a 24-hour cycle, because the pole gives weeks of low sun and no night. Here the night is made.



08 · IMMERSIVE · NIGHT SKY

Immersive · night sky.

The sky the pole never shows: a slow projection of what is above, and a reason to stay up.



08 · LOUNGE

Lounge.

Soft structure inside hard shell. The liner carries pressure; the block carries the regolith. Light comes from the seams.



08 · THE CIRCLE

The circle.

Stools in a ring around a low platform: the meeting place for the whole crew, under a ceiling that is only light and fold.



08 · QUIET ROOM

Quiet room.

A room with no function: for meditation, religion, grief or silence. Every station that lasted found it needed one.



08 · ALCOVES

Alcoves.

Three bays off a skylight: places to sit apart without leaving the room. Variety is the cure for monotony.



08 · PASSAGES

Passages.

The lattice cells become a sequence of rooms and openings: routes with choices, so the inside is never one corridor.



CONCEPT RENDER · MOONWALK 2.0

CHAPTER 09 · MOONWALK 2.0 · REPORT 03

Circadian light

The pole has no day. The room makes one.

Nothing outside tells the body when to sleep, so the light inside has to: morning, midday, evening and night in the same room, with only the light changing.

09 · CIRCADIAN RHYTHM

Morning, midday, evening, night

One room, one liner, four lights. Concept renders.



09 · MORNING

The pole has no day. The room makes one.

At the south pole the Sun circles the horizon for weeks and then is gone for weeks. Nothing outside tells the body when to sleep, so the light inside has to. Morning: warm, low, rising from the seams.



09 · MIDDAY

Midday.

Cool and bright, high colour temperature, the ceiling at its strongest. The same room, the same liner: only the light has changed, and with it the hour.



09 · EVENING

Evening.

Warm and red, light dropping to the floor. Melatonin is left alone; the room tells the crew the day is ending before the clock does.

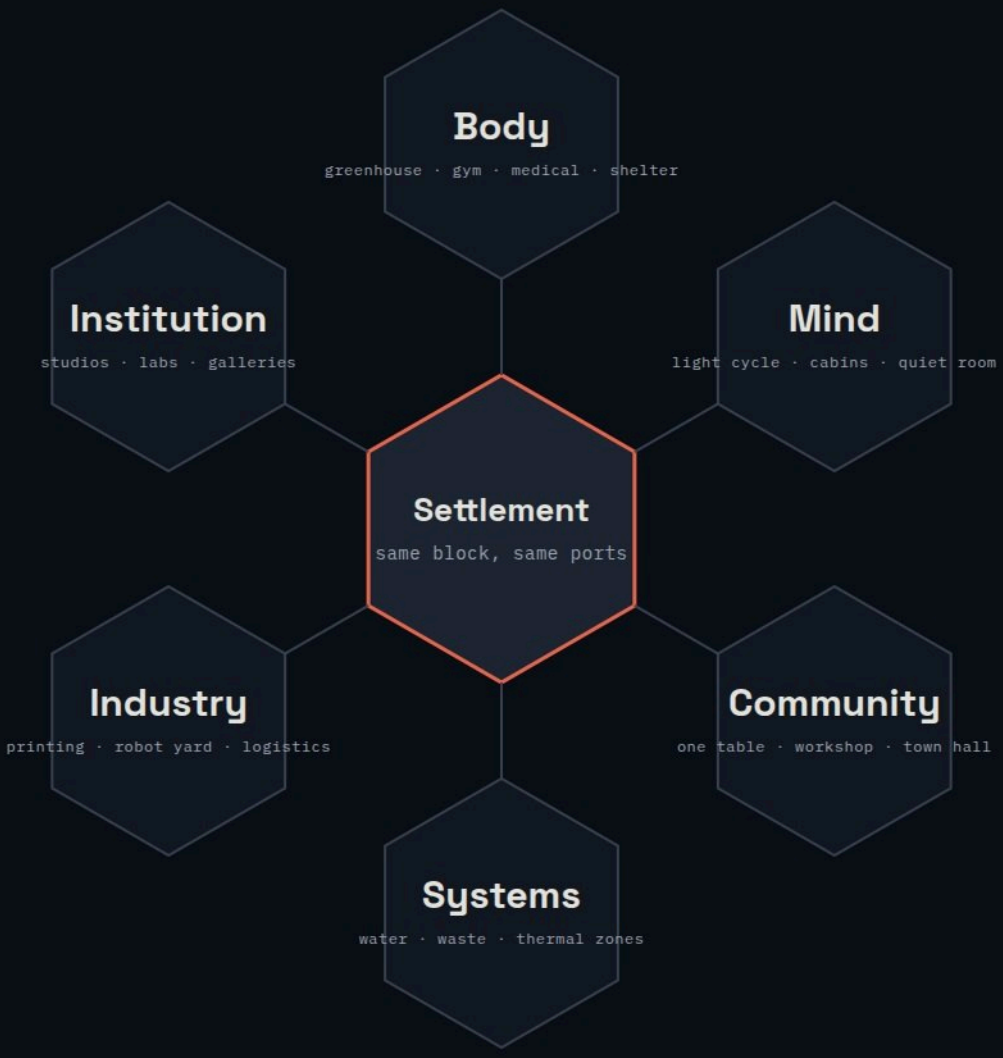


09 · NIGHT

Night.

Almost dark: a seam of light at the thresholds and the porthole, enough to move by. Polar winters taught this first: circadian disruption is the first thing to go wrong, and tunable light is the cheapest fix.

10



CHAPTER 10 · MOONWALK 2.0 · REPORT 03

Site notes and programme

What a settlement needs that a base does not

Analog stations in Antarctica, submarines and the ISS agree on what keeps people well in a sealed place. All of it is shells, seats and ports from the same block; the difference is what goes behind the port.

07 · SITE NOTES

Four candidate regions, one chosen for the walk

Connecting Ridge, de Gerlache Rim 2, Malapert Massif and Nobile Rim 1 scored in Report 02 on light, ice distance, slope and landing access. The structures in this report are drawn on Connecting Ridge.

Map tool

The site view links to JSMARS, a zero-install planetary GIS in the browser by Prof. Anwar Sounny-Slitine. It is in beta; the Moon basemap and layers are still being added.

What the site has to give

A flat, compacted shoulder for the pad; an approach below 20° for surface vehicles; a berm line between pad and vault; and foundation seats levelled to under 1° for element placement.

What NASA has not decided

Site preparation. The Moon Base ground rules (Oct 2026) say compaction and levelling are “not yet resolved” and ask bidders to state their assumptions. The pavement in this report is that assumption, made concrete.

08 · PROGRAMME

What a settlement needs that a base does not

All of it is shells, seats and ports from the same block; the difference is what goes behind the port.

BODY

Greenhouse and food; gym and a long straight run; medical bay with an isolation room; a deep radiation shelter (one thickened vault); dust control with a clean / dirty line through the whole plan.

MIND

Tunable light on a 24-hour cycle; a private cabin each and quiet rooms; Earth-facing windows and varied routes; one room with no function at all.

COMMUNITY

One table big enough for everyone and a real kitchen; a workshop open to residents; an arts and science institution that doubles as the town hall; private, good links home.

SYSTEMS

Water and waste loops with visible reserves; thermal zoning rather than one climate; a bathhouse.

INDUSTRY

Block printing and laser forming at the regolith face; a maintenance yard for the robot fleet; logistics bays rated for repeated docking.

INSTITUTION

The cultural wing: the same lattice with studios, labs and galleries behind the ports, and the Earth in every window.

NASA figures from NextSTEP-3 Appendix B (80MSFC26R0003) Amendment 2 and Directed Call 2 GRACs.



01
Isaac Sim
paving loop, wall and column re-run, first dome course

02
Environment build
real terrain and structures as one scene

03
VR walk
the settlement as a VR experience

04
Robots among people
pavers beside EVA crew, handover zones

05
Operations and interfaces
pick, haul, place, inspect; operator views

06
Geometry
tower lattice, panel seat, hybrid skin

07
Site
JSMARS region, seat tolerance vs 3°

08
Programme
zones on the masterplan

■ REPORT 04 · SIMULATION AND OPERATIONS FIRST

CHAPTER 11 · MOONWALK 2.0 · REPORT 03

Next steps

Towards Report 04

Simulation and operations first: the paving loop in Isaac Sim, the real site as one environment, a VR walk, robots among people, and the operator interfaces on top.

09 · NEXT STEPS

Report 04

Eight steps for the next report, from simulation and operations to geometry, site and programme.

01

Isaac Sim

Paving loop with two pavers and a maintainer; the wall and column re-run at the current tilt; a first dome course on the rim ring.

02

Environment build

The real site terrain and the built structures assembled in Isaac Sim as one environment, exteriors first, then the pressurised interiors, so every later run happens on the actual ground.

03

VR walk

That environment exported as a VR experience of the whole settlement: walk the pavement, enter a shell, stand in the alcove light.

04

Robots among people

Human–robot interaction staged in the same scenes: paver traffic beside EVA crew, handover zones, approach and stop distances at the ports.

05

Operations and interfaces

Robot operations simulated end to end, pick, haul, place, inspect, and the operator interfaces drafted on top: task queue, block registry, camera and map views.

06

Geometry

Tower lattice from Column A; cell-to-panel seat; hybrid skin as one Grasshopper definition from pavement to dome.

07

Site

Candidate region drawn in JSMARS once the Moon layers land; foundation seat tolerance against NASA's 3° threshold.

08

Programme

Zones on the masterplan: habitat chain, greenhouse and water, body and mind, industry, institution.



The SPACE ROBOTICS LAB

Moonwalk 2.0

REPORT 03 · FROM ONE BLOCK TO A SETTLEMENT

LIVE REPORT
arhimatrix.github.io/arhimatrix-moonwalk-2-rep_03

INTERACTIVE PARTS
3D Moon tour with site data · block to dome explorer · five end-effectors · scroll sequences

REPORTS
Report 01 · 02 · 03 · 04 at arhimatrix.github.io

IMAGES
LROC: NASA/GSFC/Arizona State University · renders are concept images · NASA figures from NextSTEP-3 Appendix B

NINA VELIMIROVIĆ · SPACE ROBOTICS LABORATORY, TOHOKU UNIVERSITY · OCTOBER 2026